AllergenTrace – installation and user manual

AllergenTrace is a web application for cafés, bakeries and small restaurants. Allergens are entered once per ingredient; the app calculates the declaration of every dish from its recipe, including sauces and doughs used inside dishes and "may contain" warnings from shared equipment. Guests read the result in a QR menu on their own phone.

This manual explains three ways to run the application and then how to use it in each role.

Way	Best for	What you need
A. Cloud version	Trying the app straight away	A web browser
B. Docker Compose	Running your own copy on a laptop	Docker Desktop (Windows, macOS) or Docker Engine (Linux)
C. Without Docker	Development	Python 3.13 and PostgreSQL 17

A. Use the cloud version

1. Open https://allergentrace.onrender.com.
2. The app runs on a free plan that sleeps after 15 minutes without visitors. The first page can then take up to one minute to load; after that, pages open normally.
3. Either log in to the demo café (the login data is given in the portfolio submission), or click **Register your business** and create your own business. A confirmation link is sent to the e-mail address you enter.

The demo café has three accounts with the same password, one per role: `demo-owner@example.com`, `demo-kitchen@example.com` and `demo-service@example.com`. These addresses cannot receive e-mails, so "Forgot your password?" does not work for them. After five wrong passwords an account is paused for 15 minutes; the other two accounts can still be used.

B. Install with Docker Compose

Step 1. Install Docker Desktop from https://www.docker.com/products/docker-desktop/ (Windows, macOS) or Docker Engine with the Compose plugin (Linux), and start it.

Step 2. Get the code, either with Git or by downloading the ZIP file from GitHub and unpacking it:

```
git clone https://github.com/sakshish12/allergentrace.git
cd allergentrace
```

Step 3. Start the app and its database:

```
docker compose up --build
```

The first start takes a few minutes, because Docker downloads Python and PostgreSQL. The app is ready when the terminal shows `Listening at: http://0.0.0.0:8000`.

Step 4. Open http://localhost:8000. To check that the app and the database work, open http://localhost:8000/health/; it shows `{"status": "ok", "database": "ok"}`.

Step 5. Register a business. Locally no e-mails are sent: the confirmation e-mail, including its link, is printed in the terminal where Docker runs. Copy the link into the browser to activate the account.

Optional: start with the demo café. Set a password before starting, and the start script creates "Café Demo" with example data and the three demo accounts:

```
DJANGO_DEMO_PASSWORD=Demo-2026 docker compose up --build             # macOS, Linux
$env:DJANGO_DEMO_PASSWORD="Demo-2026"; docker compose up --build     # Windows PowerShell
```

Stop and reset. Press `Ctrl+C` to stop. `docker compose down` removes the containers but keeps the data; `docker compose down -v` also deletes the local database.

C. Install without Docker

Step 1. Install Python 3.13 and PostgreSQL 17, and create an empty database, for example `allergentrace`.

Step 2. The PDF allergen list uses WeasyPrint, which needs the Pango library of the operating system; see https://doc.courtbouillon.org/weasyprint/stable/first_steps.html.

Step 3. In the project folder:

```
python -m venv .venv
source .venv/bin/activate          # Windows: .venv\Scripts\activate
pip install -r requirements-dev.txt
export DJANGO_DEBUG=True           # Windows: set DJANGO_DEBUG=True
export DATABASE_URL=postgres://USER:PASSWORD@localhost:5432/allergentrace
python manage.py migrate
python manage.py createcachetable
python manage.py runserver
```

Step 4. Open http://localhost:8000. Confirmation e-mails are printed in the terminal, as with Docker.

Configuration

All settings are environment variables; `.env.example` lists them.

Variable	Meaning
`DATABASE_URL`	Connection to PostgreSQL
`DJANGO_SECRET_KEY`	Long random string; required when `DJANGO_DEBUG` is not `True`
`DJANGO_DEBUG`	`True` only for local development
`DJANGO_ALLOWED_HOSTS`	Host names the app answers to, separated by commas
`BREVO_API_KEY`, `DJANGO_DEFAULT_FROM_EMAIL`	Sending e-mails through Brevo; without a key, e-mails are printed to the log
`DJANGO_CONTACT_EMAIL`	Contact address in the privacy policy and terms of use
`DJANGO_DEMO_PASSWORD`	Creates the demo café at start-up; empty means no demo café

Tests

```
pytest                                  # automated tests (needs DATABASE_URL)
ruff check . && ruff format --check .   # code style
```

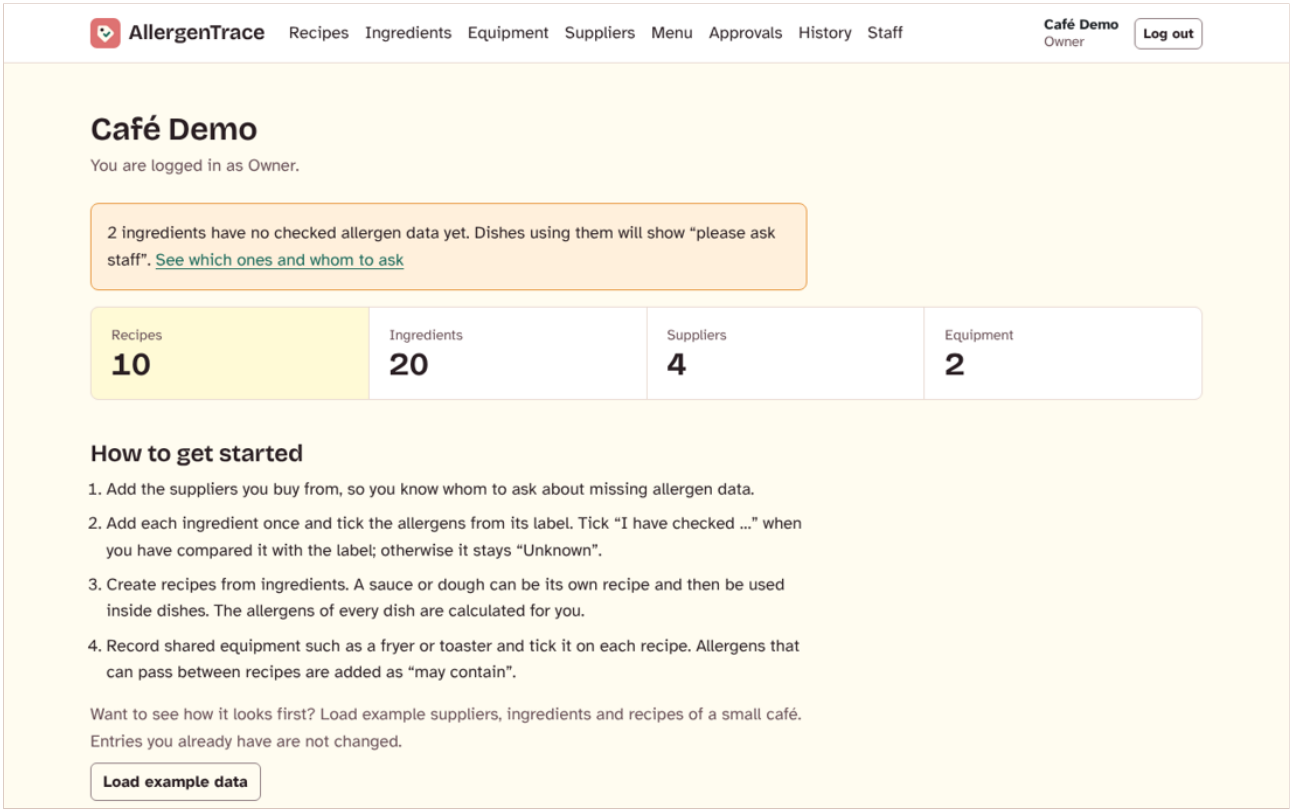
User guide

The app knows three roles. Each role includes the rights of the one below it.

Role	Can do
Owner	Everything, plus staff accounts, menu, approving allergen removals
Kitchen	Enter and change suppliers, ingredients, equipment and recipes
Service	Read everything, to answer guests' questions

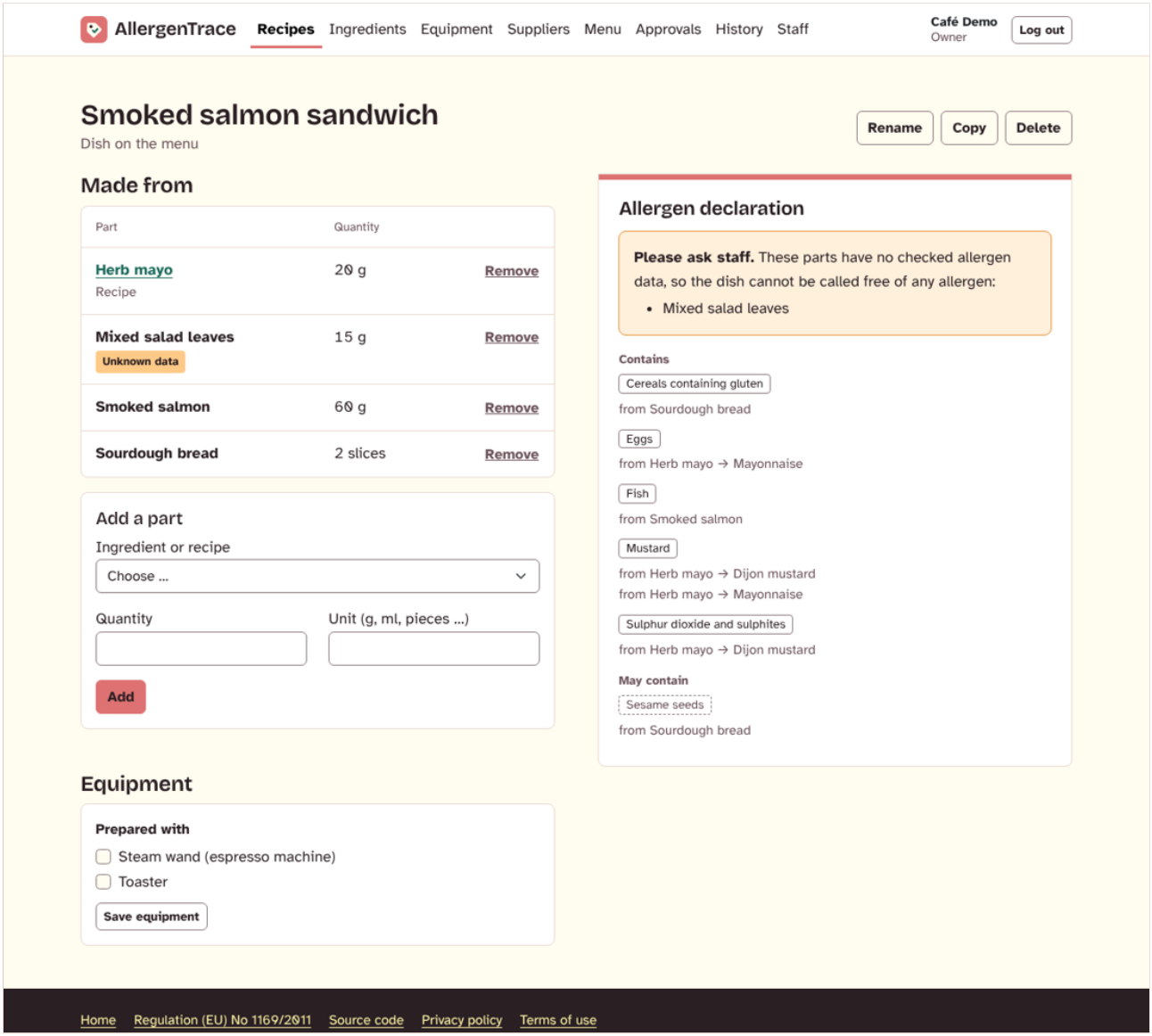
Owner: getting started

1. **Register** the business on the start page and open the confirmation link from the e-mail.
2. On the **Overview** page, click **Load example data** to see a small café with ten recipes, or start with your own data. Example entries never change entries you already have.
3. Under **Staff**, create accounts for kitchen and service staff. Accounts are deactivated instead of deleted, so the change history keeps showing who changed what.



Kitchen: ingredients and recipes

1. **Suppliers**: record whom you buy from, so you know whom to call about missing allergen data.
2. **Ingredients**: enter each ingredient once and tick its allergens from the product label, as "Contains" or "May contain (traces)". Tick **"I have checked these allergens..."** only after comparing with the label; otherwise the ingredient stays **Unknown**, and every dish using it shows "Please ask staff". **Ingredients → Unknown data** lists them with the supplier to contact.
3. **Equipment**: record shared equipment such as a toaster or fryer. Tick "dedicated to one recipe" only if it is never used for anything else.
4. **Recipes**: create a recipe, tick "Served as a dish" for dishes on the menu (untick it for sauces and doughs), and add its parts under **Add a part**. A sauce or dough can be its own recipe and be used inside dishes. Under **Equipment**, tick what the recipe is prepared with.
5. The **Allergen declaration** on the recipe page is calculated automatically and explains every allergen, for example "Milk: Pesto → Parmesan". Allergens that can pass between recipes through shared equipment appear as "May contain". **Copy** creates a variant of a recipe.



Owner: approving removals

When a change would remove an allergen from a dish, guests keep seeing the old, safer declaration until the owner approves it. New allergens are shown to guests at once. The owner receives an e-mail when something waits for approval; **Approvals** lists the changes, and **History** shows every change with time and person.

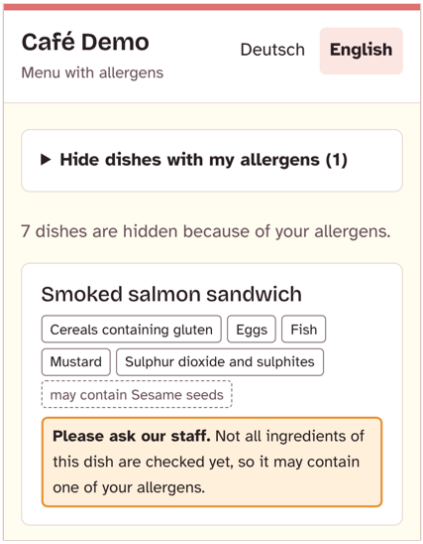
Owner: menu, QR code and allergen list

1. **Menu**: untick dishes that are not served at the moment and click **Save menu**.
2. Download the **QR code** (PNG or SVG) and print it on table cards. It opens the guest menu without a login.
3. **Download allergen list (PDF)** gives the written allergen information (14 allergens per dish) for the counter and for food control.

Guests: the QR menu

1. Scan the QR code with the phone camera; the menu opens in the phone's language (German or English) and can be switched at the top.

2. Tap **Hide dishes with my allergens**, tick what you cannot eat and tap **Show dishes**. Dishes that may contain traces are hidden as well, unless you tick "Also show dishes that may contain traces".
3. A dish with **"Please ask our staff"** uses an ingredient whose allergen data has not been checked; ask before ordering.



Troubleshooting

Problem	Solution
The cloud version takes a minute to open	The free server is waking up; wait, then it is fast.
"Too many requests. Please wait a minute"	More than 10 login or registration attempts per minute came from your network; wait one minute.
Signing in is paused	After five wrong passwords the account is paused for 15 minutes; use "Forgot your password?" to set a new one.
No confirmation e-mail when running locally	Locally, e-mails are printed in the terminal instead of being sent.
Port 8000 is already in use	Stop the other program, or change `"8000:8000"` to `"8080:8000"` in `docker-compose.yml` and open http://localhost:8080.
"Please ask staff" on a dish	One of its ingredients is "Unknown", or the recipe has no parts yet. See **Ingredients → Unknown data**.